Business Problem Statement

A leading retail company seeks to gain deeper insights into customer shopping behavior to enhance sales performance, improve customer satisfaction, and strengthen long-term loyalty. The management team has observed notable shifts in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in identifying the key drivers of consumer decisions and repeat purchases, such as discounts, product reviews, seasonal trends, and payment preferences.

The overarching business question is:

"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Project Deliverables

1. **Data Preparation & Modeling (Python)**
    - Clean, transform, and structure the raw dataset for analysis.
    - Develop models to support customer segmentation and behavioral insights.
2. **Data Analysis (SQL)**
    - Organize the dataset into a structured format.
    - Simulate business transactions and run queries to extract insights on customer segments, loyalty, and purchase drivers.
3. **Visualization & Insights (Power BI)**
    - Build an interactive dashboard highlighting key patterns and trends.
    - Enable stakeholders to explore data dynamically and make informed, data-driven decisions.
4. **Report and Presentation**
    - Prepare a comprehensive project report summarizing findings, insights, and business recommendations.
    - Deliver a professional presentation that visually communicates actionable strategies to stakeholders.

5. **GitHub Repository**
    - Maintain a well-structured repository containing Python scripts, SQL queries, and Power BI dashboard files.
    - Ensure reproducibility and transparency for future reference and collaboration.